Milad Bafarassat

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Website LinkedIn GitHub Google Scholar Istanbul

Education

Sabanci University
BSc Double Major in Electronics Engineering, Minor in Physics, Istanbul, Turkey — *Sep 2022 – Jan 2026*

Sabanci University
BSc in Computer Science and Engineering, Istanbul, Turkey — *Sep 2021 – Sep 2025*
CGPA: 3.72/4.0 – Ranked 63/729 (top 8%) in Faculty of Engineering
Awarded the *Dilek Sabanci Scholarship* for exceptional academic performance.

Sharif University of Technology — **Tehran, Iran**
Master of Business Administration (MBA) — *Feb 2016 – Feb 2019*
Ranked 5th in Masters Entrance Exam out of 30,000 students (Top 0.02%).

Sharif University of Technology — **Tehran, Iran**
BSc in Chemical Engineering — *Sep 2009 – Jul 2015*
Ranked 1374th in the Undergraduate Entrance Exam out of 308,875 students (Top 0.4%).

Publications and Patents

[C1] C. Li, R. Yang, T. Li, **Milad Bafarassat**, K. Sharifi, Dirk Bergemann, Zhuoran Yang. (2024)
STRIDE: A Tool-Assisted LLM Agent Framework for Strategic and Interactive Decision-Making
AutoRL Workshop at **ICML 2024**

[C2] Milad Bafarassat, Melik Yazici, Korkut Kaan Tokgoz. (2025)
FET Modeling with Deep Neural Networks and GAN-Augmented Small Measurement Dataset
Oral Presentation at **SMACD 2025**

[P1] Milad Bafarassat, Korkut Kaan Tokgoz. (2025)
A Deep Neural Network-Based FET Simulation Method and System Implementing the Model
Application No.: 2025/008782 **Filing Date**: June 30, 2025 **Reference No.**: P7069-TR

Research Experience

Researcher, Sabanci University
AI for Wireless Communications, Advisor: Prof. Özgür Gürbüz — *Sep 2025 – Present*
Developed LSTM-based predictive models to estimate low-latency drone state variables (orientation and angular rates) from limited IMU and RF telemetry streams to optimize antenna array beamforming and per-drone antenna gain patterns. Work includes time-series feature engineering, sequence modeling (stacked LSTMs and attention mechanisms), domain-aware loss design for orientation estimation, and integration of model outputs into antenna design simulations to improve link budget and coverage.

Researcher, Sabanci University
Backdoor Attacks on Time-Series Models, Advisor: Prof. Emre Özfatura — *Sep 2025 – Present*
Research focuses on security evaluation and attack development for state-of-the-art time-series models. Implemented and extended two recent approaches: (1) input-aware poisoning attacks that craft sample-specific triggers, and (2) generative-model based trigger synthesis (GAN/conditional generative models) to produce stealthy backdoors against RNNs, LSTMs and transformer-based time-series classifiers. Conducted threat-model design, empirical evaluations on standard benchmarks, and initial defenses/mitigations.

Remote Researcher, Shanghai AI Lab
Mechanistic Analysis of Language Models, Advisors: Dr. Jie Fu, Wenyu Du (HKU PhD) — *Jan 2025 – Mar 2025*
Researched mechanistic analysis and interpretability of transformer-based language models to enhance AI reliability and safety. Engaged with cutting-edge advancements in mechanistic interpretability, critically reviewed state-of-the-art research papers, and implemented a Meta-CoT-based approach to automate mechanistic interpretability in transformer architectures.

Graduation Project, Sabanci University
FET Modeling using Deep Learning, Advisor: Prof. Korkut Kaan Tokgoz — *Feb 2024 – Jun 2025*
Focused on developing robust FET models using data-driven techniques. This research explored data augmentation strategies, including interpolation methods, Generative Adversarial Networks (GANs), and Autoencoders, to enhance a limited dataset of 80,000 samples from 27 transistors. Following augmentation, deep learning models are trained and refined for predictions. The project aimed to improve semiconductor device modeling and integrate these models into CAD tools such as SPICE or Verilog-A. Project concluded in paper accepted for Oral Presentation at **SMACD 2025**.

Remote Researcher, Yale University
LLM Agent Framework for Strategic Decision-Making, Advisors: Prof. Zhuoran Yang, Dr. Li — *Oct 2023 – May 2024*
Conducted research on enhancing decision-making by LLMs in multi-agent settings by integrating external tools. Paper was accepted to the ***AutoRL Workshop at ICML 2024*** and ***INFORMS Workshop on Market Design at EC 2024***.

Researcher, Sabanci University
Determining Political Affiliations of Twitter Users using CV and ML, Advisor: Prof. Onur Varol — *Jul 2023 – Oct 2023*
Implemented advanced Self Supervised Learning clustering algorithms, including SwAV and BARLOW Twins, to categorize over 7 million profile pictures effectively. Then, clusters were used to determine the distribution of political affiliations and human/bot ratios.

Teaching Experience

Programming Fundamentals (CS201) — **Sabanci University**
Learning Assistant — *Spring 2024*

Internships

Intraverse — **Lugano, Switzerland (Remote)**
AI Engineering Intern — *Jun 2025 – Jul 2025*
Designed and implemented a middleware pipeline connecting Unity game engine events to LLM APIs, enabling dynamic NPC dialogue, decision-making, and skill systems. Developed a plugin-based architecture for extensible character and quest design, with blockchain-backed validation for NPC responses. Produced documentation, a functional prototype, and performance benchmarks for real-time multiplayer scenarios.

Work Experience

HezarDastan Holding — **Istanbul, Turkey (Remote)**
AI Product Manager — *Feb 2020 – Mar 2023*
Led a cross-functional team of 7 software engineers and ML researchers, and 2 operations and marketing team members. Co-founded and introduced 3 innovative AI-driven products, expanding the company's product portfolio. Spearheaded the development and implementation of **Recrupen**, an in-house Applicant Tracking System (ATS), streamlining hiring across 6 subsidiaries and processing 5000+ resumes annually to facilitate over 200 hires. Launched **QShoot**, an AI-based math problem-solving app, achieving 10K users within 2 months of launch. Launched **Gorgeous**, an AI-powered makeup recommendation app using user photos and event context; reached 1K users in 2 days.

HezarDastan Holding — **Tehran, Iran**
Human Resources Manager — *Sep 2015 – Jan 2020*
Managed a team of 40 professionals across HR, Recruitment, Learning & Development, and Administration. Built and scaled a recruitment team from scratch to 7 full-time recruiters. Drove company growth by increasing headcount from 100 to 300 within two years, with a focus on technical roles. Promoted to HR Manager within 3 years due to consistent performance and leadership. Improved employee satisfaction significantly, raising the Net Promoter Score (NPS) from low 80%s to 92%.

Extracurricular Activities

Program for Undergraduate Research (PURE)
Mentor — *Jan 2025 – Jun 2025*
Mentored junior students on a project focused on leveraging reinforcement learning techniques to enhance transistor data augmentation and optimize learning outcomes.

Sabanci AI Club (kAi)
Student Advisor — *Apr 2023 – Oct 2023*
Helped in the identification and negotiation processes with potential sponsors for the club.

Select Coursework

Graduate-level courses: Internet of Things Sensing System, Scalable Learning Systems
Advanced courses: Probability and Statistics (A), Linear Algebra (A), Algorithms and Data Structures (A-), Quantum Mechanics I (A-), Quantum Mechanics II (B+), Intro to Signal Processing (A), Advanced Programming (A), Computer Networks (B+), Multimedia Communication (A), Information and Coding Theory (B+), Statistical Modeling (A), Data Science (A), Computer Vision (A-), Digital Image & Video Analysis (A-)

Skills

Software Skills: C++, Python, PyTorch, TensorFlow
Languages: English (C1), Turkish (C1), Azeri (Native), and Persian (Native)

Tests and Scores

GRE (323): Quantitative: 170, Verbal: 153
TOEFL iBT (100): (R24, L29, S21, W26)